$$p_\theta(b_{t,j} \mid I_{t-5:t}, e)$$

$$\mathcal{L} = -\sum_{t,j} \log p_\theta(b^*_{t,j} \mid I_{t-5:t}, e)$$

$$-\log 0.2 \approx 1.609, \quad -\log 0.8 \approx 0.223$$